CS341 · COMPUTER GRAPHICS

Calculations تدريب

تدريب على كل المسائل الحسابية

محارب

01 الأسئلة

ASPECT RATIO **Q1**

a) A screen is 12 inches wide and 9 inches high with a resolution of 1024 × 768. Find the aspect ratio.

b) A screen is 10 inches wide and 8 inches high with a resolution of 800 × 600. Find the aspect ratio.

RESOLUTION **Q2**

A monitor measures 32 cm × 24 cm and shows 1280 × 960 pixels. Find the resolution in pixels per cm in x and y, and the aspect ratio.

FRAME BUFFER **Q3**

How much frame-buffer memory (in bytes) does a 1024 × 768 screen need with:
a) 1 bit per pixel b) 8 bits per pixel c) 24 bits per pixel

VECTORS **Q4**

a) P = (2, 5), Q = (7, 1). Find the vector from P to Q and its length.

b) a = (1, 2, 2). Find |a| and the unit vector â.

c) b = (2, −1, 0). Find a · b. Are a and b perpendicular?

d) Find a × b.

e) Find the angle between u = (1, 0) and w = (−1, 1).

f) Find the counterclockwise perpendicular of c = (3, −2).

POINTS & LINES **Q5**

a) Which are valid points? $0.5P + 0.5R$, $2P - R$, $P + R$, $0.2P + 0.3R$

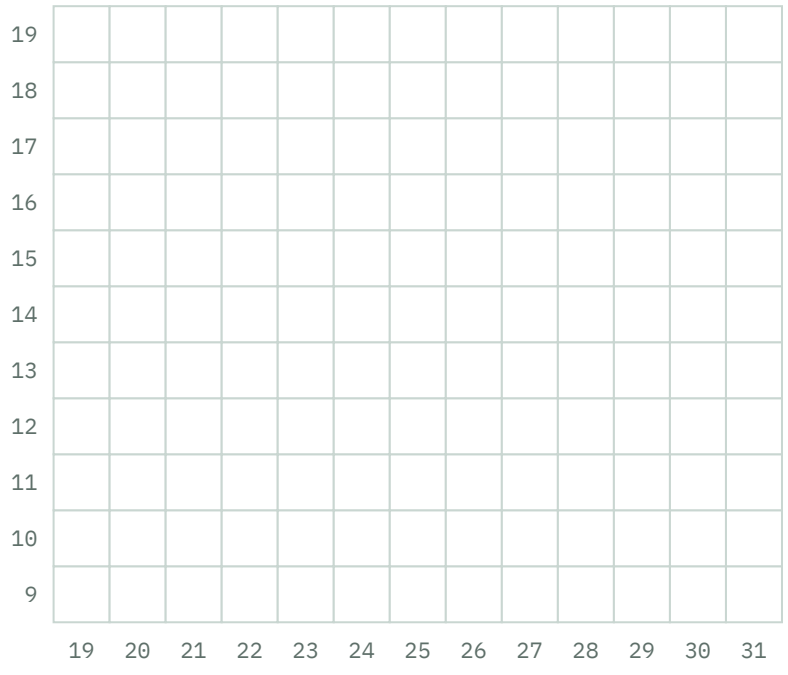
b) $A = (1, 2)$, $B = (9, 6)$. Find the point at $t = 0.75$ on $P = A + t(B - A)$.

c) For $C = (2, 1)$ and $B = (5, 5)$, write the parametric form and the point-normal form of the line.

DDA **Q6**

Using DDA, find the pixels from (20, 10) to (30, 18).

k	x	y	round(x)	round(y)
0				
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				



DDA **Q7**

Using DDA, find the pixels from (1, 1) to (7, 4).

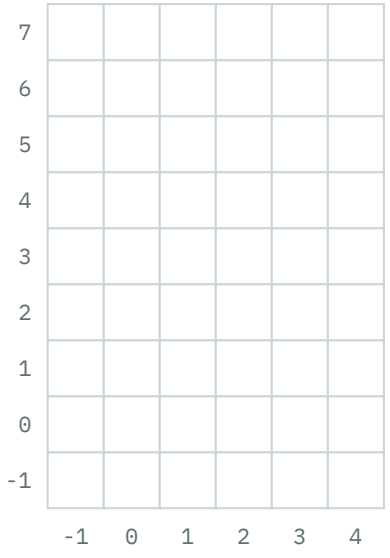
k	x	y	round(x)	round(y)
0				
1				
2				
3				
4				
5				
6				



DDA · |m| > 1 **Q8**

Using DDA, find the pixels from (0, 0) to (3, 6).

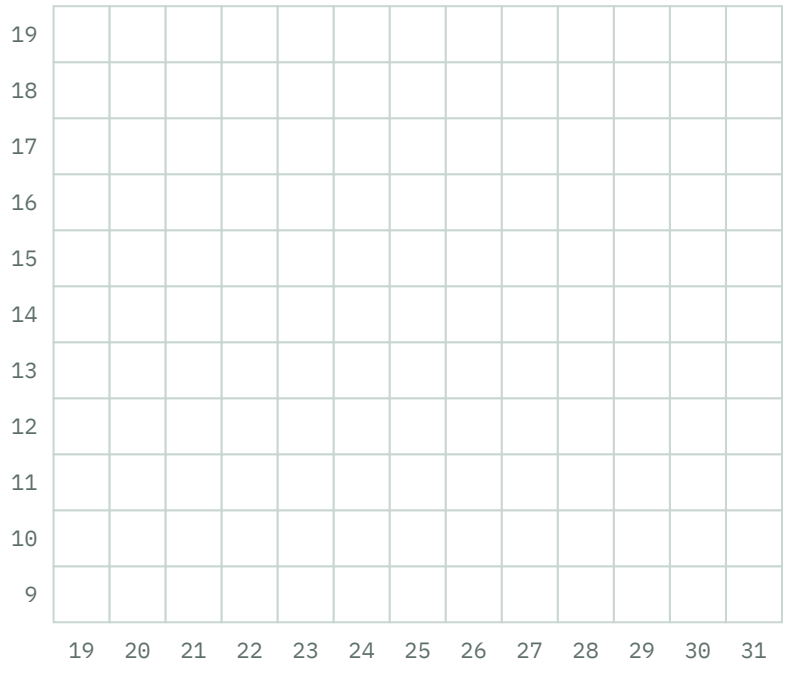
k	x	y	round(x)	round(y)
0				
1				
2				
3				
4				
5				
6				



BRESENHAM **Q9**

Using Bresenham's algorithm, find the pixels from (20, 10) to (30, 18).

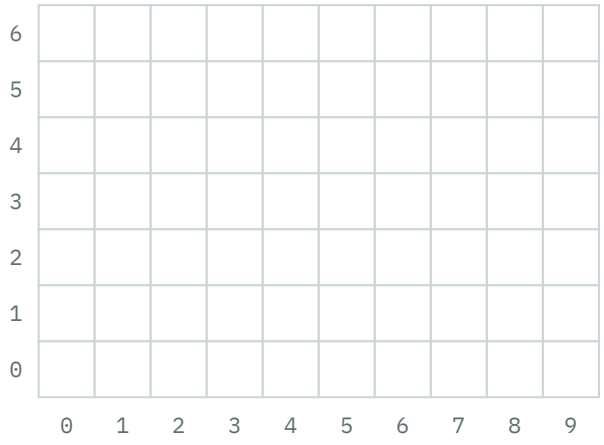
k	p_k	القرار	البكسل التالي	p_{k+1}
0				
1				
2				
3				
4				
5				
6				
7				
8				
9				



BRESENHAM **Q10**

Using Bresenham's algorithm, find the pixels from (1, 1) to (8, 5).

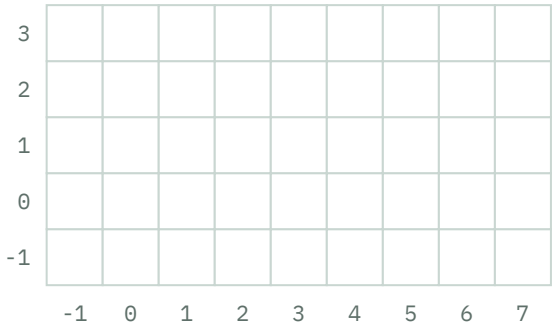
k	p_k	القرار	البكسل التالي	p_{k+1}
0				
1				
2				
3				
4				
5				
6				



BRESENHAM **Q11**

Using Bresenham's algorithm, find the pixels from (0, 0) to (6, 2).

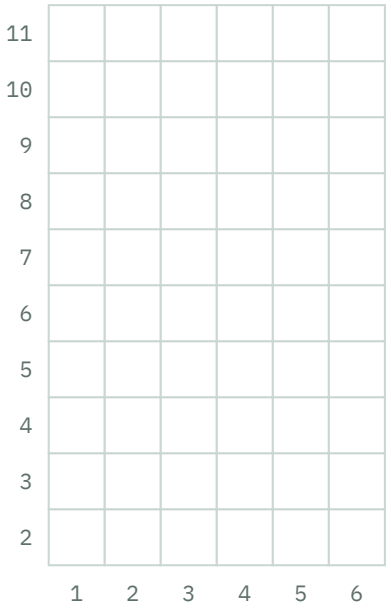
k	p_k	القرار	البكسل التالي	p_{k+1}
0				
1				
2				
3				
4				
5				



BRESENHAM · m > 1 **Q12**

Using Bresenham's algorithm, find the pixels from (2, 3) to (5, 10).

k	p_k	القرار	البكسل التالي	p_{k+1}
0				
1				
2				
3				
4				
5				
6				



CIRCLE **Q13**

Using Bresenham's circle algorithm, find the pixels of the first (top) octant of a circle centered at the origin with r = 8.

CIRCLE **Q14**

Using Bresenham's circle algorithm, find the pixels of the first (top) octant for r = 10, then write the 8 symmetric points of (3, 10).